

Introduction to L^AT_EX

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at
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What is $\text{\LaTeX}$? [1]

- It is a macro package based on $\text{\TeX}$ [2].
- Its purpose is to simplify $\text{\TeX}$ typesetting, especially for documents containing mathematical formulae.
- Its document processing is essentially programming: you create a text file in $\text{\LaTeX}$ markup and the $\text{\LaTeX}$ compiler reads this in order to produce the final document.
- It is widely used in academia.

A First Document in $\text{\LaTeX}$: Source Code

```
\documentclass[12pt]{article}
\usepackage[british]{babel}

\newcommand{\Editor}{\textbf{Emacs}} % define macros!

\title{A Sample \LaTeX\ Report} % used by \maketitle
\author{Pierre Flener} % used by \maketitle
\date{\today} % used by \maketitle

\begin{document}

\maketitle % generates the title page

\section{Revisiting  $n$ -Queens}
I typed this file with a plain-text editor.
(I used \Editor.)

\end{document}
```

A First Document in $\text{\LaTeX}$: Result of Compilation

A Sample $\text{\LaTeX}$ Report

Pierre Flener

27th May 2025

1 Revisiting n -Queens

I typed this file with a plain-text editor. (I used **Emacs**.)

A Table in $\text{\LaTeX}$: Source Code (from the Demo Report)

```
\begin{tabular}{rrrrrrrrrrrrrrrr} % right [r] for decimal-point alignment
  \input{res-M4CO.tex}
\end{tabular}
```


A Table in $\text{\LaTeX}$: Result of Compilation

Backend	Chuffed		CP-SAT		Gecode		Gurobi		PicatSAT		Yuck	
n	obj	time	obj	time	obj	time	obj	time	obj	time	obj	time
3	20	1	20	585	20	606	20	684	20	693	20	t/o
4	34	180	34	696	34	646	34	1123	34	764	34	t/o
5	26	381910	26	2594	26	47128	26	65629	26	t/o	26	t/o
6	–	t/o	27	t/o	–	t/o	26	296077	28	t/o	26	t/o

Skeleton Reports

- The course website has $\text{\LaTeX}$ source files with the imposed structure for the assignment reports and the project report, and with indications on how to proceed.
Use them! This saves your time and our time.
Our demo report exemplifies the expected quality of content.
- Our `run_backends.sh` script conducts the experiments and generates a result table that is automatically imported (rather than manually copied) into your $\text{\LaTeX}$ report, so each time you change the model, it suffices to re-run that script and re-compile your report, without any tedious number copying!
- You focus on the **content** in order to generate a professional-looking document **without** thinking about the formatting.
- We will **not** accept reports spread over multiple PDF files: write into separate files and use `\input{filename}`.
- You can share-edit using Overleaf.

Online Resources

- Download $\text{\LaTeX}$: <https://www.latex-project.org/get>
- General help: <https://en.wikibooks.org/wiki/LaTeX>
- Detexify: $\text{\LaTeX}$ handwritten symbol recognition
<https://detexify.kirelabs.org/classify.html>
- Two of the best websites to find answers to $\text{\LaTeX}$ questions are
<https://tex.stackexchange.com> and
<https://stackoverflow.com/questions>
- Share editing: <https://www.overleaf.com>

References

- [1] LaTeX. $\text{\LaTeX}$. <https://en.wikipedia.org/wiki/LaTeX>.
- [2] TeX. $\text{\TeX}$. <https://en.wikipedia.org/wiki/TeX>.

Questions?

